

Pharmaceutical Reform and Pharmacy Entry

Model Results and Parallel-Trend Diagnostics

Wenxia Chen

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Hospital DDD: Pharmacy Count

	1km		2km		3km	
	id+year	prov-year	id+year	prov-year	id+year	prov-year
Public $\times$ Post	-0.403* (0.192)	-0.343* (0.158)	-1.272** (0.491)	-0.773 (0.403)	-2.148* (0.856)	-0.819 (0.724)
Post policy	1.081*** (0.148)		4.171*** (0.430)		8.757*** (0.814)	
Hospital FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No	Yes	No
Province-year FE	No	Yes	No	Yes	No	Yes
NTL control	Yes	Yes	Yes	Yes	Yes	Yes
N	164,674	164,663	164,674	164,663	164,674	164,663
R ²	0.895	0.906	0.906	0.918	0.911	0.922

Notes: Standard errors are clustered at the county level. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Hospital Results Across Outcome Models

	Count OLS			ln(count+1)			ln(count)			PPML Count		
	1km	2km	3km	1km	2km	3km	1km	2km	3km	1km	2km	3km
Public $\times$ Post	-0.343* (0.158)	-0.773 (0.403)	-0.819 (0.724)	-0.014 (0.016)	-0.022 (0.017)	-0.032 (0.018)	-0.014 (0.015)	0.010 (0.015)	-0.002 (0.015)	-0.036* (0.015)	-0.029* (0.013)	-0.027* (0.012)
Hospital FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
NTL control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	164,663	164,663	164,663	164,663	164,663	164,663	134,776	146,998	151,432	147,711	156,544	159,753

Notes: OLS columns use hospital FE and province-year FE. PPML coefficients are semi-elasticity style estimates.

Hospital TWFE Log Outcomes

	1km		2km		3km	
	id+year	prov-year	id+year	prov-year	id+year	prov-year
$\ln(\text{count}+1)$	-0.002 (0.018)	-0.014 (0.016)	-0.004 (0.020)	-0.022 (0.017)	-0.010 (0.021)	-0.032 (0.018)
$\ln(\text{count})$	-0.005 (0.017)	-0.014 (0.015)	0.025 (0.017)	0.010 (0.015)	0.018 (0.017)	-0.002 (0.015)
Post policy	Yes	Yes	Yes	Yes	Yes	Yes
Hospital FE	Yes	Yes	Yes	Yes	Yes	Yes
NTL control	Yes	Yes	Yes	Yes	Yes	Yes

Entries are coefficients on $\text{Public} \times \text{Post}$. Standard errors are clustered at the county level.

Hospital CSDID Average ATT

	1km	2km	3km
Count	-0.004 (0.169)	-0.447 (0.433)	-1.127 (0.755)
$\ln(\text{count}+1)$	0.013 (0.017)	0.013 (0.018)	0.008 (0.019)
$\ln(\text{count})$	-0.009 (0.017)	0.004 (0.016)	-0.006 (0.015)

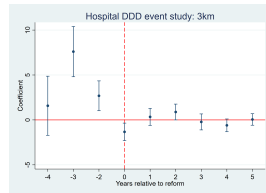
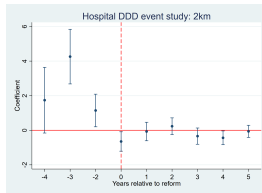
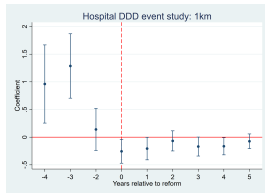
CSDID uses public hospitals as treated in their local reform year and non-public hospitals as never treated. Controls include post-policy status and NTL.

Hospital Extensive Margin

	1km	2km	3km
Public $\times$ Post	-0.002 (0.007)	-0.007 (0.007)	-0.008 (0.006)
NTL	0.037*** (0.006)	0.052*** (0.007)	0.058*** (0.008)
Hospital FE	Yes	Yes	Yes
Province-year FE	Yes	Yes	Yes
N	164,663	164,663	164,663
R ²	0.803	0.773	0.741

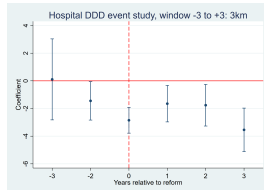
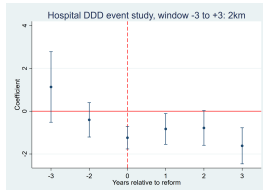
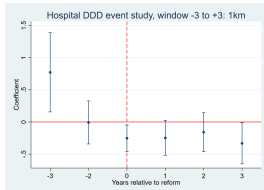
Outcome is an indicator for having at least one pharmacy within the distance band.

Hospital TWFE Event Study: Count

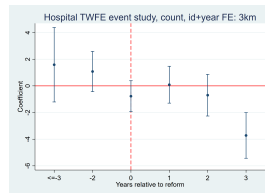
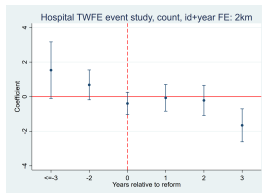
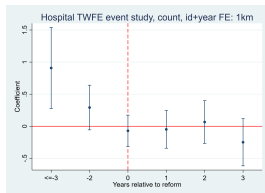


Event-time coefficients are interactions between public hospitals and relative reform years. Relative year -1 is omitted.

Hospital TWFE Event Study: Count, Window -3 to +3

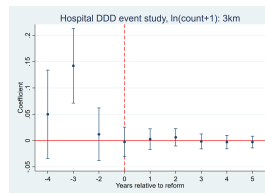
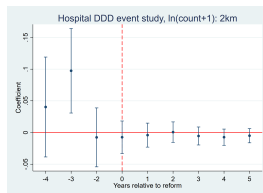
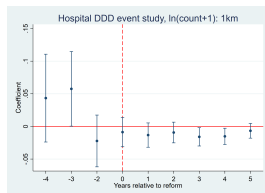


Hospital TWFE Event Study: Count, ID + Year FE

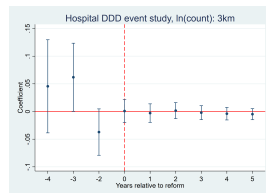
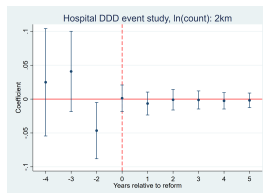
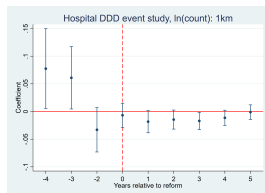


Window is restricted to event time $[-3, 3]$; relative year -1 is omitted. Models absorb hospital and year fixed effects.

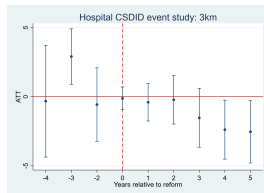
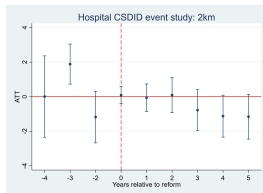
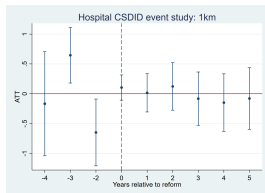
Hospital TWFE Event Study: $\ln(\text{count}+1)$



Hospital TWFE Event Study: $\ln(\text{count})$

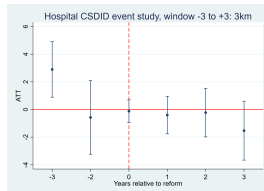
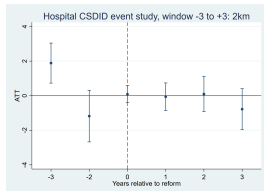
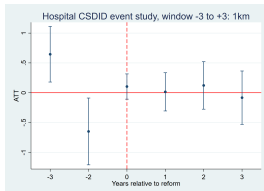


Hospital CSDID Event Study: Count

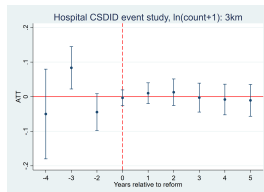
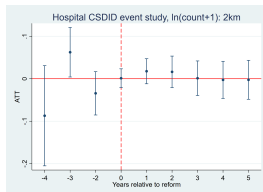
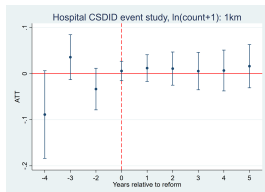


CSDID treats public hospitals in their local reform year and codes non-public hospitals as never treated.

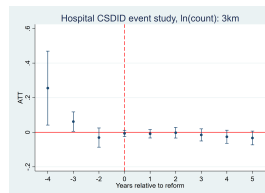
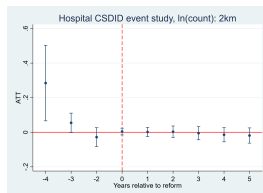
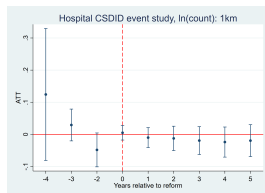
Hospital CSDID Event Study: Count, Window -3 to +3



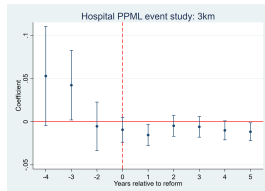
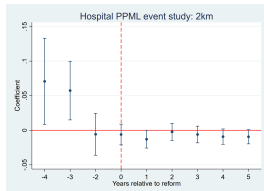
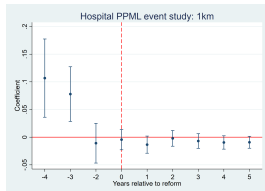
Hospital CSDID Event Study: $\ln(\text{count}+1)$



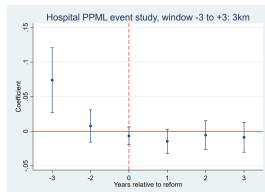
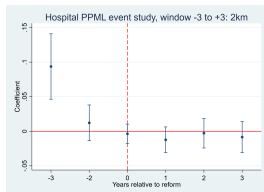
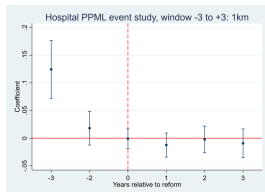
Hospital CSDID Event Study: In(count)



Hospital PPML Event Study: Count



Hospital PPML Event Study: Count, Window -3 to +3



County TWFE DID: Pharmacy Count

	FE	+ controls	+ prov-year	2012–2020 no road
Post policy	7.652*** (1.474)	5.449*** (1.454)	5.188*** (1.493)	5.396** (1.669)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	No
Province-year FE	No	No	Yes	Yes
Controls	No	Yes	Yes	Yes
N	16,961	16,961	16,961	19,496
R ²	0.824	0.845	0.872	0.877

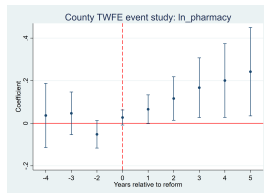
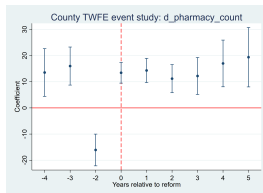
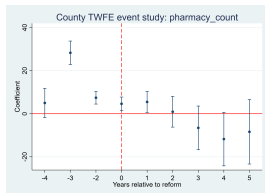
Controls include log population, log nighttime lights, and road density when available.

County Results Across Outcome Models

	Annual Change		Log Count		Has Pharmacy		PPML Count	
	FE	prov-year	FE	prov-year	FE	prov-year	FE	prov-year
Post policy	19.892*** (2.124)	19.090*** (2.123)	0.020 (0.014)	0.026 (0.016)	0.030*** (0.004)	0.025*** (0.005)	-0.014 (0.012)	-0.014 (0.015)
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province-year FE	No	Yes	No	Yes	No	Yes	No	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	14,279	14,279	15,469	15,464	16,961	16,961	16,947	16,929

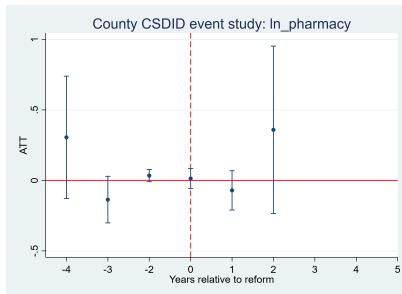
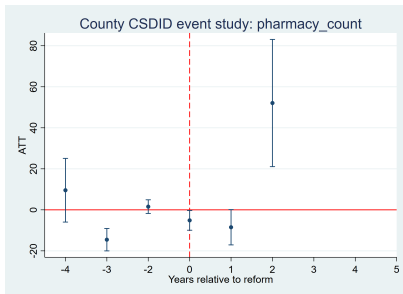
Outcome models use the controlled county sample. PPML keeps zero-count observations but yields coefficients on a different scale.

County Parallel Trends: TWFE Event Study



Panels show pharmacy count, annual change in pharmacy count, and log pharmacy count.

County Parallel Trends: CSDID Event Study



CSDID uses not-yet-treated counties as controls.

How the Results Connect

- ▶ **Hospital-level level-count estimates are negative, but log estimates are close to zero.** TWFE count and PPML estimates are negative for public hospitals after reform, while both $\ln(count + 1)$ and $\ln(count)$ TWFE coefficients are small and statistically weak.
- ▶ **Hospital CSDID results do not show a strong average effect.** CSDID average ATT estimates are negative for count outcomes but imprecise; both log specifications are near zero across 1km, 2km, and 3km buffers.
- ▶ **County-level TWFE estimates are positive in levels.** Reform exposure is associated with higher county pharmacy counts and larger annual increases, with stable positive signs after adding controls and province-year fixed effects.
- ▶ **Scale matters for interpretation.** Hospital results compare public and non-public hospitals within local reform timing, while county results compare county-level pharmacy supply before and after reform adoption.
- ▶ **Dynamic diagnostics remain the main caution.** Count-outcome event studies are shown in both regular and $[-3, 3]$ windows; log-outcome event studies are shown in the regular window only. Pre-period deviations should be reported as a limitation.
- ▶ **Overall pattern.** The reform coincides with broader county-level pharmacy expansion, but the hospital-level evidence does not support a clear relative increase in pharmacy entry around public hospitals.